



Mastering Cloud Automation

The Power 🦾 of Infrastructure as Code (IaC)

(by Stefan Rapp, 09th December 2025, 16:00 – 16:45)



HashiCorp

Terraform

```
resource "azurerm_role_assignment" "before" {
  for_each
  name
  description
  scope
  role_definition_id
  role_definition_name
  principal_id
  principal_type
  condition
  condition_version
  delegated_managed_identity_resource_id
  skip_service_principal_aad_check

  = var.role_assignments_before != null ? var.role_assignments_before : {}
  = each.key
  = each.value.description
  = try(var.role_assignment_scope_ids[each.value.scope_name], each.value.scope_name)
  = each.value.role_definition_id
  = each.value.role_definition_name
  = try(local.role_assignment_possible_principal_ids[each.value.principal_id], each.value.principal_id)
  = each.value.principal_type
  = each.value.condition
  = each.value.condition_version
  = each.value.delegated_managed_identity_resource_id
  = each.value.skip_service_principal_aad_check

  months ago | 1 author

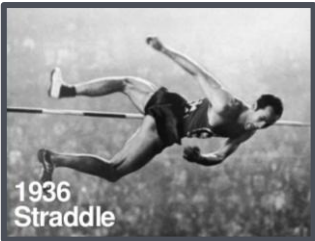
  resource "azurerm_machine_learning_workspace" "this" {
    = [azurerm_role_assignment.before]
    = var.create_machine_learning_workspace ? 1 : 0
    = var.machine_learning_workspace_name
    = try(azurerm_resource_group.this[0].name, var.resource_group_name)
    = var.module_location
    = var.machine_learning_workspace_application_insights_id
    = var.machine_learning_workspace_key_vault_id
    = var.machine_learning_workspace_storage_account_id
    = var.machine_learning_workspace_container_registry_id
    = var.machine_learning_workspace_public_network_access_enabled
    = var.machine_learning_workspace_image_build_compute_name
    = var.machine_learning_workspace_description
    = var.machine_learning_workspace_friendly_name
    = var.machine_learning_workspace_high_business_impact
    = var.machine_learning_workspace_possible_identity_ids[var.machine_learning_workspace_v1_legacy_mode_enabled]
    = var.machine_learning_workspace_sku_name

    != null ? var.machine_learning_workspace_encryptions : {}

    key_vault_id, encryption.value.key_vault_id
    id, encryption.value.key_id != null ? encryption.value.key_id : null
    possible_identity_ids[encryption.value.use
```

Olympic High Jump

Motto: “*Faster, Higher & Stronger*”



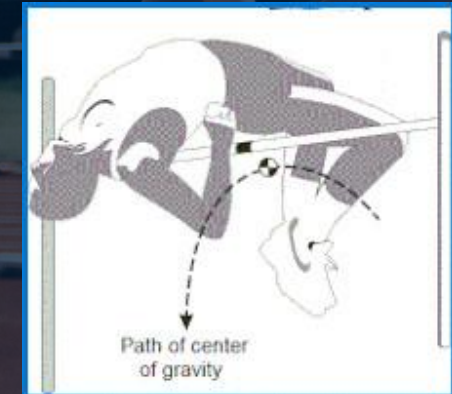
? Techniques

- › Standing Jump
- › Scissors
- › Straddle
- › Western Roll



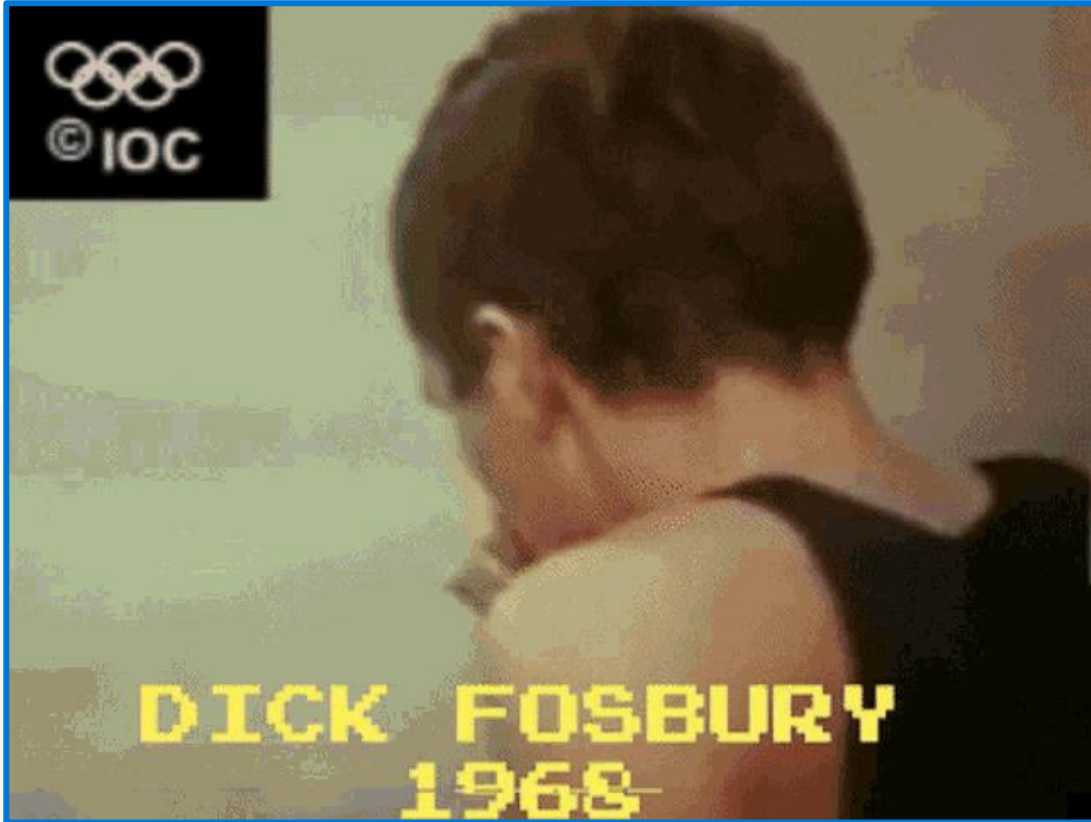
Fosbury Flop

- › High Jump + Engineering Know-How = **Center of gravity**
- › In 1968, Mexico City at the **Olympic Games**
- › Newspapers wrote, ...like a “two-legged-camel”... 🐫
- › Result: 2,24 m – Gold Medal, Olympic **Record** 🏆
- › “*I think quite a few kids will begin trying it my way now.*”



Revolutionary

The “*Fosbury Flop*”!

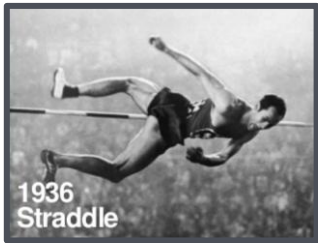


It is now the only way to do high jump! – Revolutionary!



Technology Shift/Change

Why: “Infrastructure as Code” (IaC)



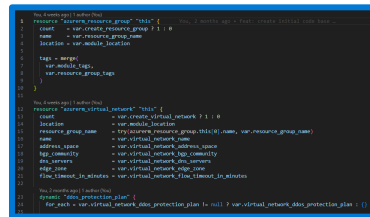
Techniques



GUI Deployment



Fosbury Flop



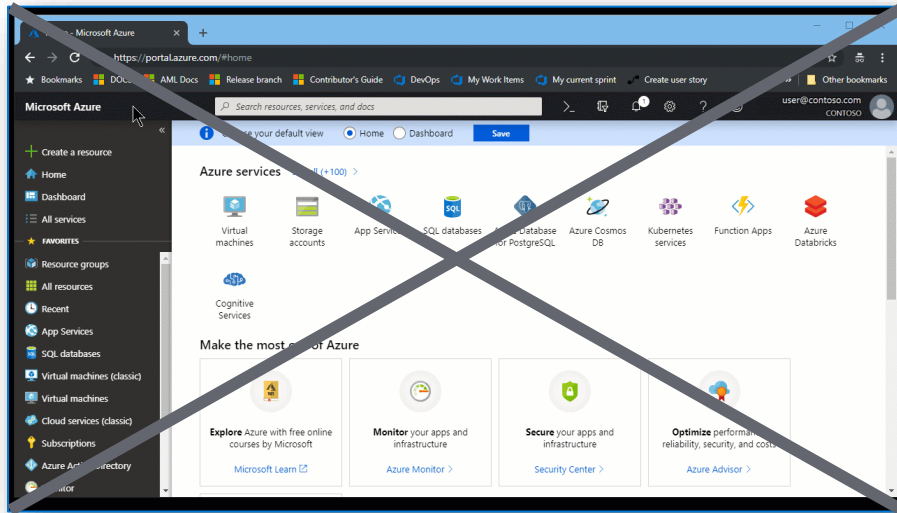
Cloud Automation



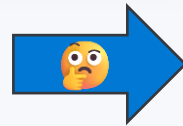


Mindset of IaC

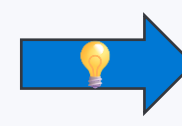
Mind change of (Cloud ☁️) resource deployment



„Click-Click-Cloud“
„Click-Ops“
„Clicky-Bunti“



Mind Change: 🧠
From “static” to
“dynamic” infrastructure



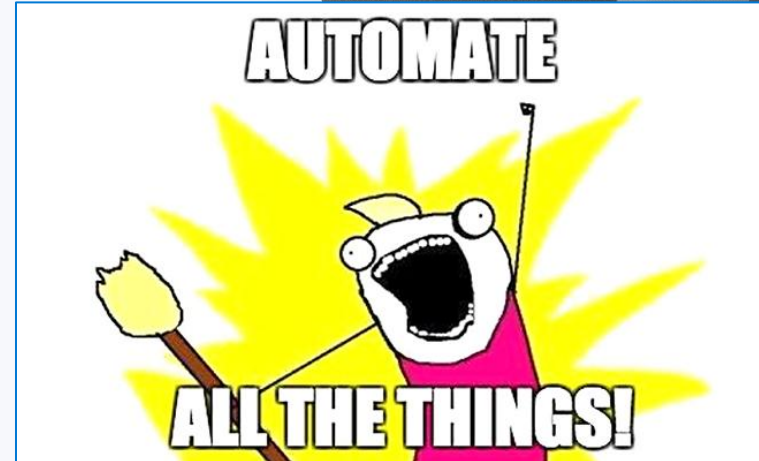
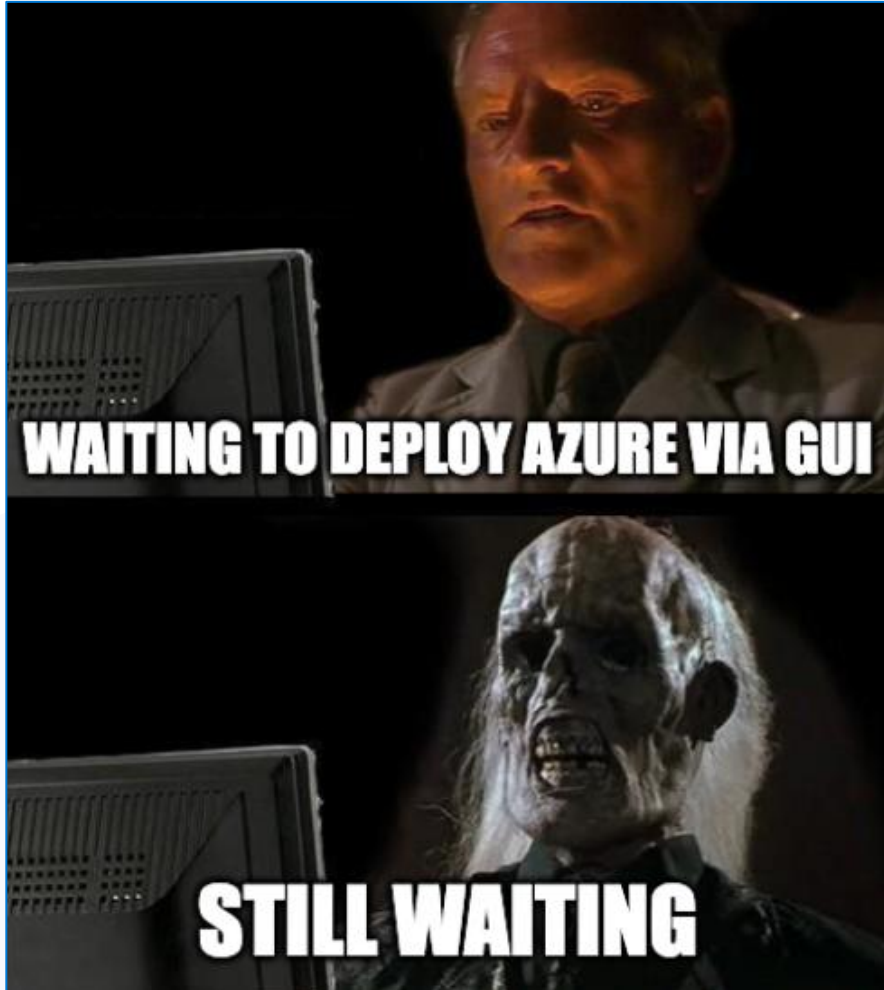
CI/CD-driven
Infrastructure deployment ⚙️



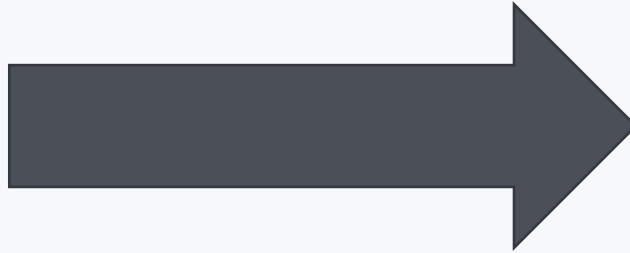
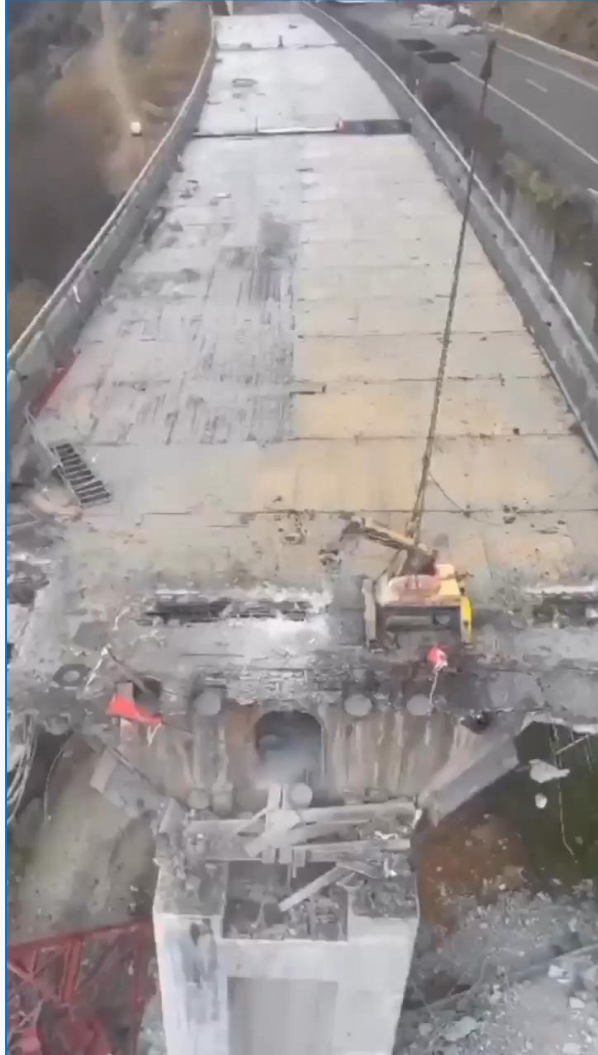
```
1 resource "azurerm_resource_group" "this" {  
2   count = var.create_resource_group ? 1 : 0  
3   name = var.resource_group_name  
4   location = var.module_location  
5  
6   tags = merge(  
7     var.module_tags,  
8     var.resource_group_tags  
9   )  
10 }  
11  
12 resource "azurerm_virtual_network" "this" {  
13   count = var.create_virtual_network ? 1 : 0  
14   location = var.module_location  
15   resource_group_name = try(azurerm_resource_group.this[0].name, var.resource_group_name)  
16   name = var.virtual_network_name  
17   address_space = var.virtual_network_address_space  
18   bgp_community = var.virtual_network_bgp_community  
19   dns_servers = var.virtual_network_dns_servers  
20   edge_zone = var.virtual_network_edge_zone  
21   flow_timeout_in_minutes = var.virtual_network_flow_timeout_in_minutes  
22  
23   dynamic "ddos_protection_plan" {  
24     for_each = var.virtual_network_ddos_protection_plan != null ? var.virtual_network_ddos_protection_plan : {}  
25   }
```



Why using IaC instead of a GUI?



Hell – How bad Cloud Automation (IaC) works?





Focus on IaC!

What is IaC?



Managing/Provisioning of (cloud) **infrastructures** using **code**.

What resources, capabilities and environments are **configured**.

Declarative syntax can be used to specify the resources and detailed settings.

Code describes the **resources** and the whole **architecture landscapes**.

Human readable code files (`.tfvars`) describe how the infrastructure **looks like**.

Definition of **variables** and their **values** of the environment (e.g., DEV, TEST, PROD).

[illegible]



How does Terraform Code look like?

DEMO Time!

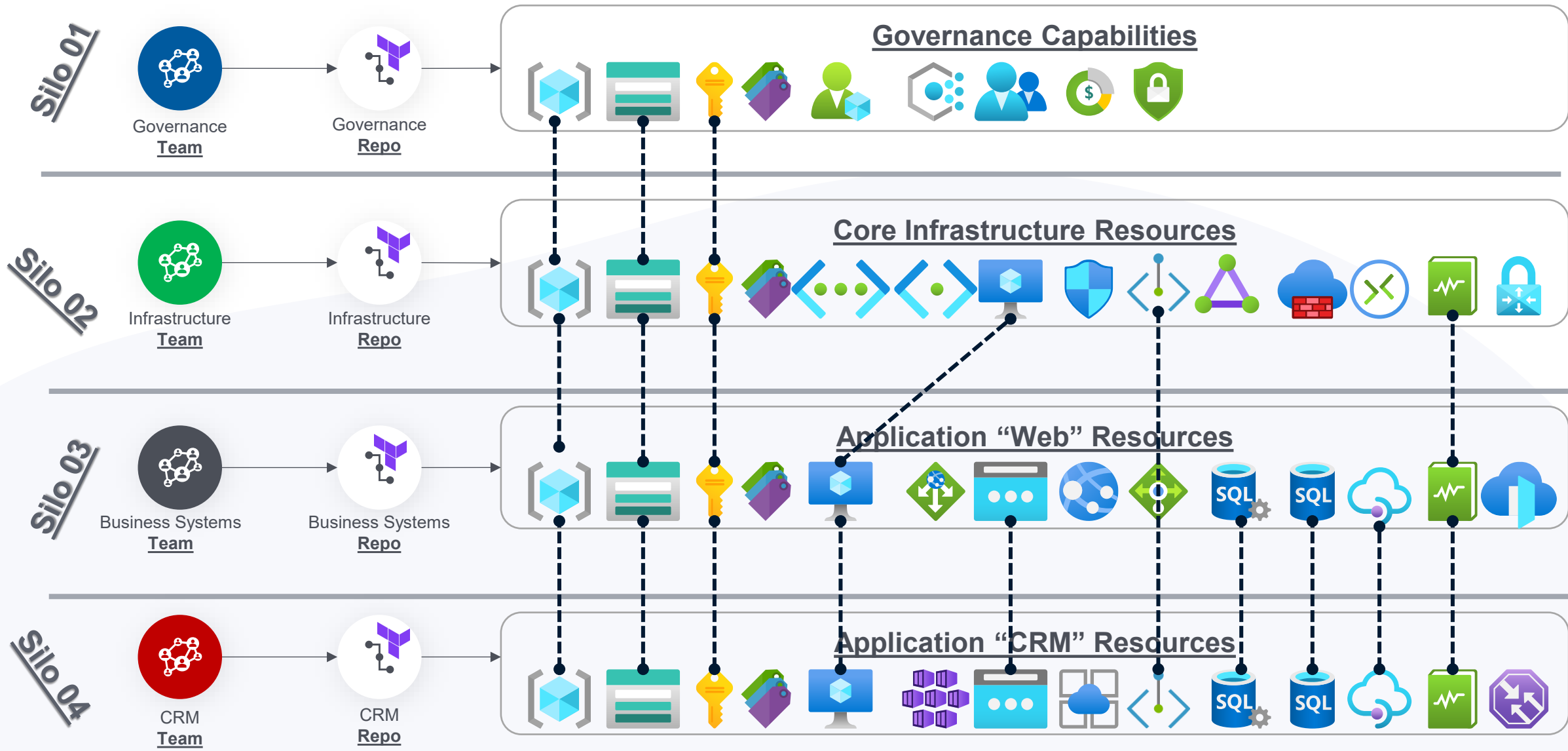
Advantages of using an IaC approach

- ✓ Reusability
 - Creation of "repeating" infrastructures (**standardization/template**).
- ✓ Speed
 - **Faster** creation of resources and environments.
- ✓ Parallelization
 - Several developers can work on the same infrastructure **simultaneously**.
- ✓ Documentation
 - The **code definition file** is also the documentation of the environment.
- ✓ Consistency
 - The code reflects the **overall state** of the environment.
- ✓ Tracking
 - Use of source code management (**version control**).
- ✓ Order
 - Order of resource creation is managed **automatically**. Automate changes.
- ✓ Error/state handling
 - **No check** necessary if resources already exist ("*Desired State*").
- ✓ Quality
 - Less **human error** by using the same code ("*no-brainer*").
- ✓ Reproducibility
 - The **same code** always leads to the **same result**.
- ✓ Cost Reduction
 - Remove **manual** tasks. People refocus their efforts on enterprise tasks.



Using Modules in Terraform

Current Situation in organization



A module in Terraform is...

the **definition** of infrastructure (“What?”). 🏛️

independently **usable** & **testable**. 🧪

reusable across different **projects** or **environments**. 🌴

separated in code repositories. 🏗️

used to **encapsulate** “complex” logic. 🧑🔧

is **easier** to manage, test, update and consume. 🔄

“Is a container for multiple resources that are used together.”

“A collection of Terraform files and folder in a repo.”

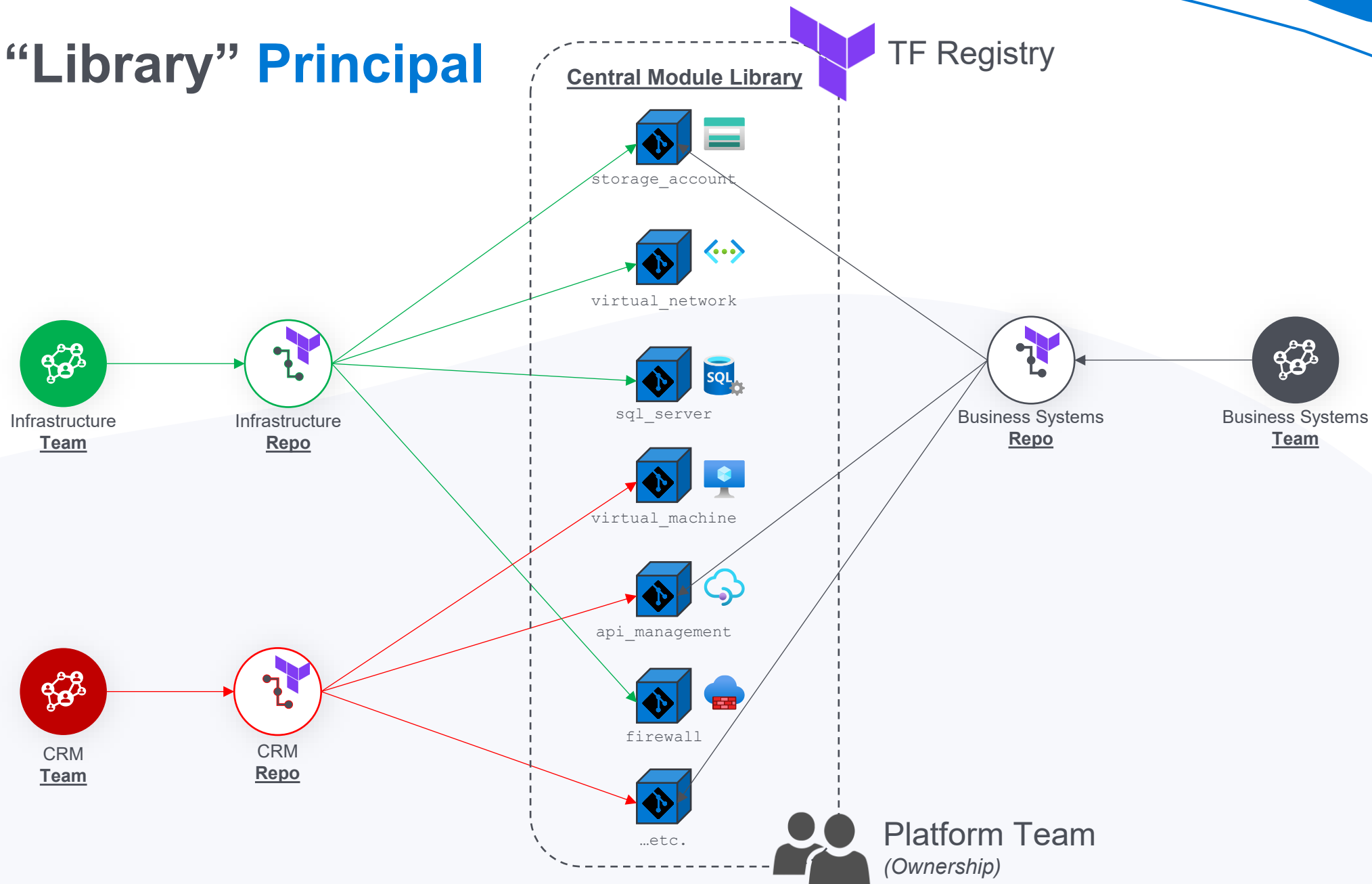
“Is the baseline for producing quality and reusable Terraform code.”



How does a Module look like in #Code?

DEMO Time!

The “Library” Principal



Standardized Module Structure

1.

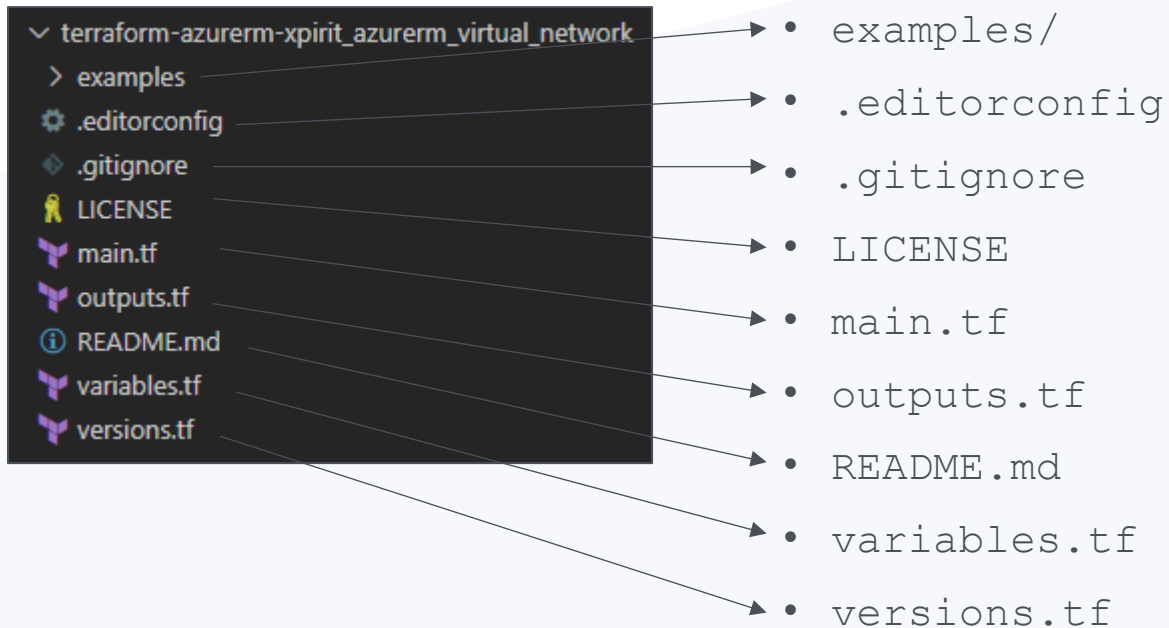
file & directory layout

2.

Distributed in separate
repositories

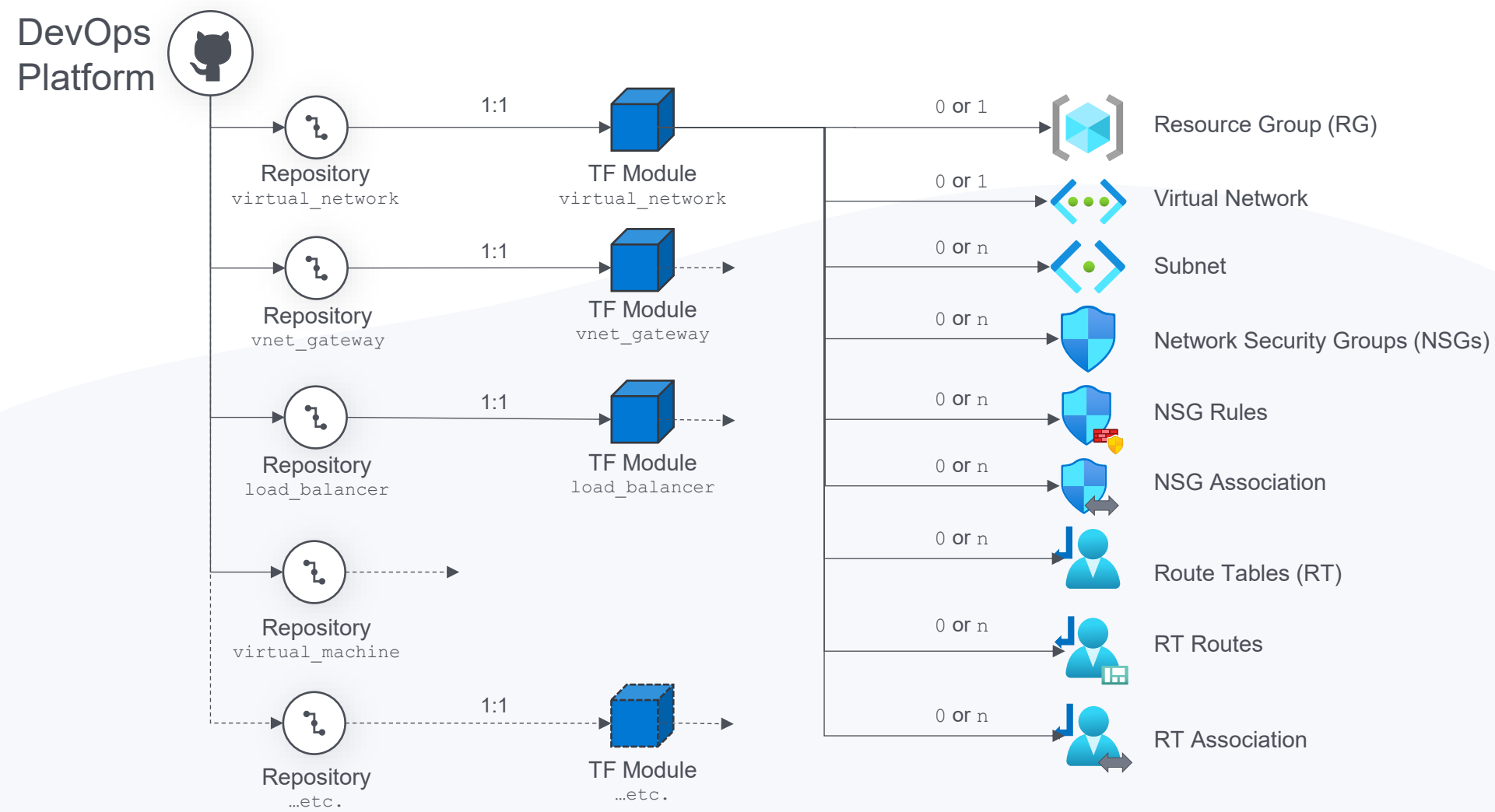
3.

Structure to generate
documentation



- separate subdirectory for each example incl. README.md
- maintain consistent code style ([Example](#))
- specifies untracked files that Git should ignore ([Example](#))
- which this module is available (publicly, privately)
- primary entry point, containing all needed resources
- to return results to the calling module
- what the module should be used for (basic documentation)
- (declaration) to accept values from the calling module.
- (constraint) define supported terraform & provider version

Standard Module Architecture



Distinguish Bad from Good Modules



Bad Modules 🤔

- ... can drive you **nuts**. 🤪
- ... are **horrible** to use. 🤢
- ... lead to bad **results**. 🤔



Good Modules 😊

- ... are **fun** to use. 🤪
- ... lead to **great** results. 👍
- ... are **easy** to maintain and manage. 🛠️

➔ Blog Post: [The **Art** of Modularization in Terraform – Mr. Azure](#) 🖼️ 🎨 🖋️



How to **manage** a module according to its lifecycle?

Publishing & Versioning

Use a Terraform Registry

TF Cloud helps teams to **provision** infrastructure.

Support for **versioning** and list of **providers** and **modules**.

Handles **downloads** and **access** with TF Cloud API tokens.

Consumers do not need **direct** access to the module's repository.

Public Registry



<https://registry.terraform.io>

Share Terraform providers and Terraform modules public.



Private Registry



<https://app.terraform.io>

Hosted on an organization's private registry and are only available to members

```
87 module_location = var.bastion_host_module_location
88 module_tags      = var.bastion_host_module_tags
89
90 # azurerm_resource_group
91 create_resource_group = var.create_bastion_host_resource_group
92 resource_group_name   = var.bastion_host_resource_group_name
93
94 # azurerm_public_ip
95 create_public_ip      = var.create_bastion_host_public_ip
96 public_ip_name        = var.bastion_host_public_ip_name
97 public_ip_domain_name_label = var.bastion_host_public_ip_name
98 public_ip_zones       = var.bastion_host_public_ip_zones
99
100 # azurerm_bastion_host
101 create_bastion_host = var.create_bastion_host
102 bastion_host_name   = var.bastion_host_name
103 # 4 weeks ago | 1 author (You)
104 bastion_host_ip_configurations = {
105   # 4 weeks ago | 1 author (You)
106   "2a8947-2efa-4b0e-ac61-c4086259a419" = {
107     public_ip_address_id = null
108     subnet_id            = module.virtual_network.subnet_ids["AzureBastion"]
109   }
110 }
111
112 # 4 weeks ago | 1 author (You)
113 private_dns_resolver = {
114   # 4 weeks ago | 1 author (You)
115   "app.terraform.io/Customer-ML-Pritix/xpirit_azurerm_private_dns_resolver" = {
116     version = "53.0-0.0.2"
117   }
118 }
119
120 private_dns_resolver_module_location = var.private_dns_resolver_module_location
121 private_dns_resolver_module_tags    = var.private_dns_resolver_module_tags
122
123 # azurerm_resource_group
124 create_resource_group = var.create_private_dns_resolver_resource_group
125 resource_group_name   = var.private_dns_resolver_resource_group_name
126
127 # azurerm_private_dns_resolver
128 create_private_dns_resolver = var.create_private_dns_resolver
129 private_dns_resolver_name   = var.private_dns_resolver_name
130 private_dns_resolver_virtual_network_id = module.virtual_network.virtual_network_id
131
132 private_dns_resolver_inbound_endpoint = var.create_private_dns_resolver_inbound_endpoint
133 private_dns_resolver_inbound_endpoints = var.private_dns_resolver_inbound_endpoints
134 private_dns_resolver_inbound_endpoint_subnets = module.virtual_network.subnet_ids["AzureBastion"]
135
136 private_dns_resolver_outbound_endpoint = var.create_private_dns_resolver_outbound_endpoint
137 private_dns_resolver_outbound_endpoints = var.private_dns_resolver_outbound_endpoints
138 private_dns_resolver_outbound_endpoint_subnets = module.virtual_network.subnet_ids["AzureBastion"]
```



How does a Registry look like?

DEMO Time!

How to use modules in Templates?

- Local file system

```
module "dd34208a_7c40_4dd2_beb5_4a5bf4d21d22" {  
  source = "../../module-library-azurerm/terraform-azurerm-xpirit_azurerm_management_group"  
  
  create_management_group      = true  
  management_group_name       = "c06835d9-1135-4b19-b403-749c06443f4c"  
  management_group_display_name = "mg-pritix"  
  management_group_subscription_association_subscription_ids = []  
}
```

- Private registry

```
module "dd34208a_7c40_4dd2_beb5_4a5bf4d21d22" {  
  source = "app.terraform.io/Customer-ML-Pritix/xpirit_azurerm_management_group/azurerm"  
  version = "3.42.0-0.0.2"  
  
  create_management_group      = true  
  management_group_name       = "c06835d9-1135-4b19-b403-749c06443f4c"  
  management_group_display_name = "mg-pritix"  
  management_group_subscription_association_subscription_ids = []  
}
```

- Directly from the repository

```
module "dd34208a_7c40_4dd2_beb5_4a5bf4d21d22" {  
  source = "git::https://github.com/rapster83/terraform-azurerm-xpirit_azurerm_management_group?ref=v3.42.0-0.0.3"  
  
  create_management_group      = true  
  management_group_name       = "c06835d9-1135-4b19-b403-749c06443f4c"  
  management_group_display_name = "mg-pritix"  
  management_group_subscription_association_subscription_ids = []  
}
```

- Public registry (→ TF Registry)

```
module "dd34208a_7c40_4dd2_beb5_4a5bf4d21d22" {  
  source = "azrappster83/management_group/azurerm"  
  version = "4.0.0"  
  
  create_management_group      = true  
  management_group_name       = "c06835d9-1135-4b19-b403-749c06443f4c"  
  management_group_display_name = "mg-pritix"  
  management_group_subscription_association_subscription_ids = []  
}
```

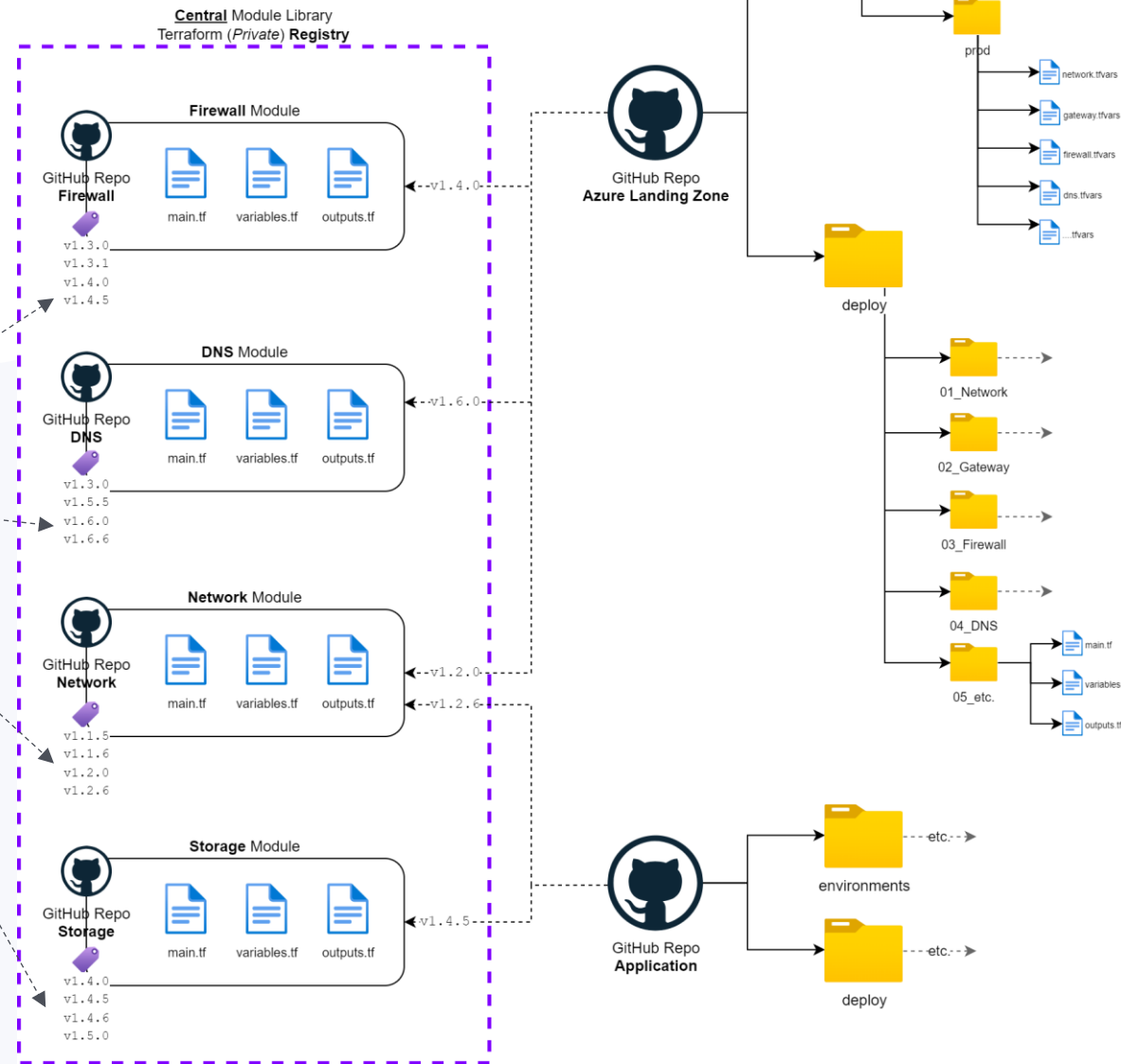
Use [SemVer](#)



Repo Structure & Environments

Reference Modules in Templates

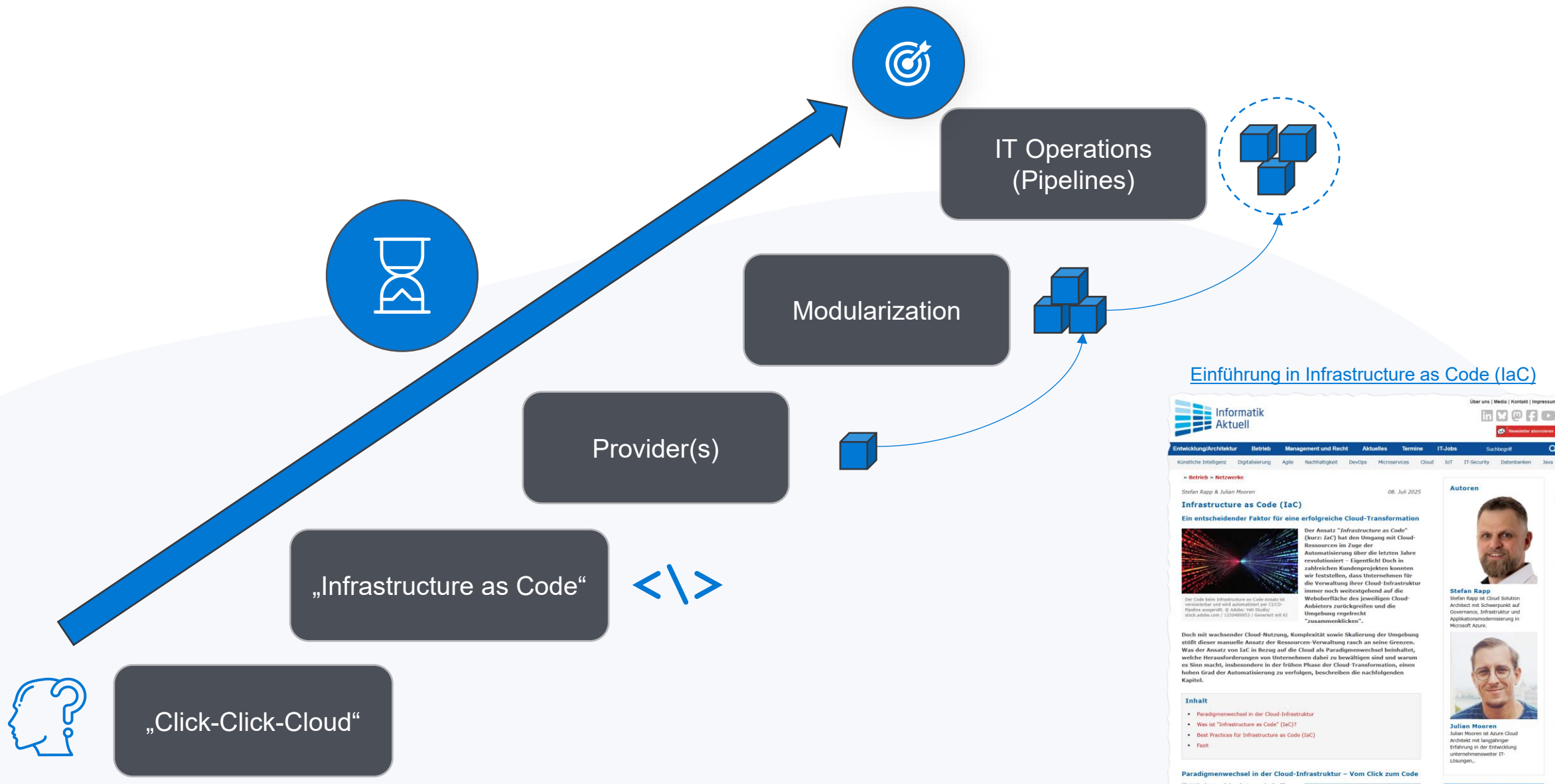
Use [SemVer](#):





Cloud Operating Model

Cloud Automation – Challenge

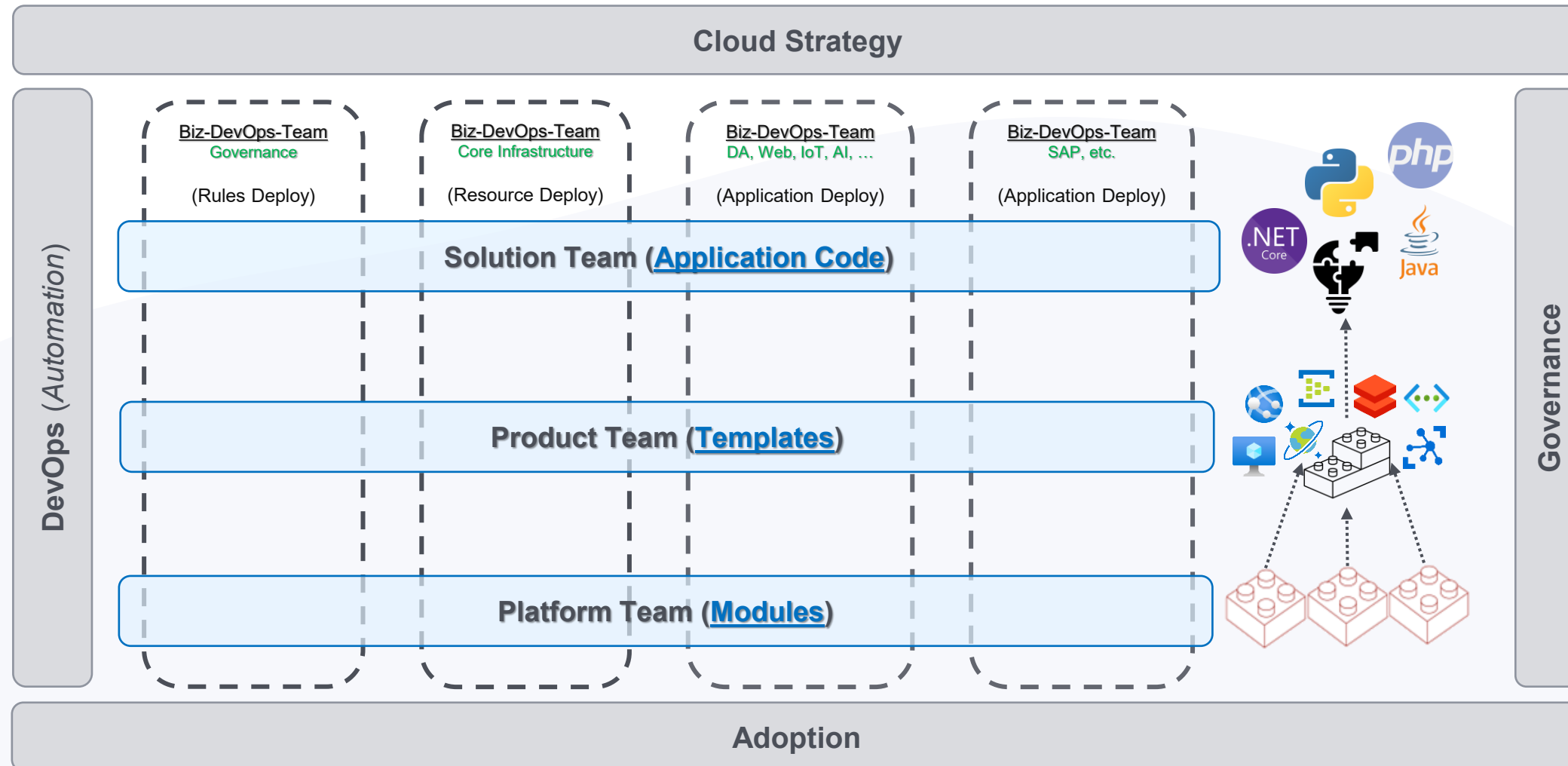


Future Cloud Operating Model

Principal:

"You deploy it, You own it"

"You build it, You run it"





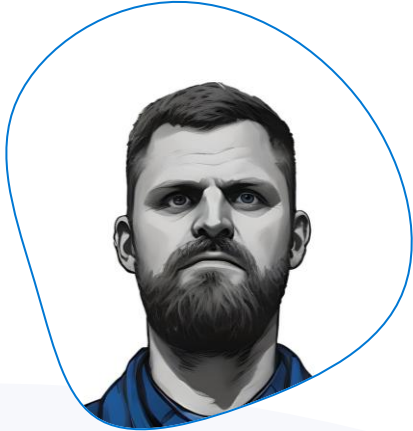
Key Takeaways

Key Takeaways of IaC

- ✓ Affects **people**, **processes**, and **technology**. 
 - ✓ Changes traditional **IT roles** and **responsibilities**.  ([Blog Post](#))
-
- ✓ Follow the code **best-practices**. 
 - ✓ Start Modularization from the **beginning**!  TOP
 - ✓ Make sure that all modules “*look the same*” (**standardization**). 
 - ✓ Use a **central** module library across projects or departments. 
 - ✓ Keep pace with the **technology** (“*Don’t get left behind!*”) 

BEST

PROFILE – Speaker



Stefan Rapp

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Xebia

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#AzureRocks 🙌 🧑💻 🎸



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